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Organization of Frontoparietal Cortex in the Tree Shrew (*Tupaia belangeri*). I. Architecture, Microelectrode Maps, and Corticospinal Connections

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ABSTRACT

Despite extensive investigation of the motor cortex of primates, little is known about the organization of motor cortex in tree shrews, one of their closest living relatives. We investigated the organization of frontoparietal cortex in Belanger's tree shrews (*Tupaia belangeri*) by using intracortical microstimulation (ICMS), corticospinal tracing, and detailed histological analysis. The results provide evidence for the subdivision of tree shrew frontoparietal cortex into seven distinct areas (five are newly identified), including two motor fields (M1 and M2) and five somatosensory fields (3a, 3b, S2, PV, and SC). The types of movements evoked in M1 and M2 were similar, but M2 required higher currents to elicit movements and had few connections to the cervical spinal cord and distinctive cyto- and immunoarchitecture. The borders between M1 and the anterior somatosensory regions (3a and 3b) were identified primarily from histological analysis, because thresholds were similar between these regions, and differences in corticospinal neuron distribution were subtle. The caudal (SC) and lateral (S2 and PV) somatosensory fields were identified based on differences in architecture and distribution of corticospinal neurons. Myelin-dense modules were identified in lateral cortex, in the expected location of the oral, forelimb, and hindlimb representations of S2, and possibly PV. Evidence for a complex primate-like array of motor fields is lacking in tree shrews, but their motor cortex shares a number of basic features with that of primates, which are not found in more distantly related species, such as rats. J. Comp. Neurol. 497:133–154, 2006.
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Indexing terms: M1; cortical; premotor; somatosensory; stimulation; maps

Tree shrews (see Fig. 1) are small mammals that superficially resemble arboreal squirrels, but are actually closely related to primates (see Fig. 2; Liu et al., 2001; Murphy et al., 2001a,b; Waddell and Kishino, 2001; Springer et al., 2003). In fact, tree shrews are the closest living relatives of primates that are available for detailed neuroanatomical and electrophysiological study. Consequently, a special emphasis has been placed on studying the nervous system of tree shrews for comparison with that of their primate relatives. Our group has been particularly interested in comparing the organization of the cerebral cortex of tree shrews with that of their primate and nonprimate relatives to determine the cortical areas that might have appeared at different stages in the mammalian or Euarchontan lineages (Kaas, 2002). Beyond driving theories of brain evolution, these comparisons provide additional support for the existence of particular character states in human cortex, because traits widely held in these lineages are likely to be found in humans (Kaas, 2002).

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Fig. 1. External morphology of the Belanger's tree shrew (*Tupaia belangeri*); note the elongated snout and large eyes. Scale bar = 5 cm. © Michael S. Remple, with full rights granted to John Wiley & Sons, Inc.

The organization of sensory cortex in tree shrews has been investigated in some detail (see, e.g., Sur et al., 1980, 1981a; Weller et al., 1987; Lyon et al., 1998; Bosking et al., 2002; Alexander et al., 2004), but relatively little is known about the organization of the tree shrew motor cortex. Previous studies have provided evidence for one (Le Gros Clark, 1924) or more (Zilles, 1978) motor areas in the frontal cortex of tree shrews, based on architectonic features such as a poorly developed layer IV and the presence of large pyramidal neurons. However, Lende (1970) was unable to identify a distinct motor field in this region based on electrophysiological parameters (using surface stimulation mapping), and he proposed that tree shrews lacked a motor cortex that was distinct from somatosensory cortex. The results of terminal degeneration (Jane et al., 1969) and corticospinal cell labeling (Nudo and Masterton, 1990) studies provided inconclusive evidence for the location and existence of one or more motor areas in tree shrews, insofar as large regions of the frontoparietal cortex appeared to project to the cervical spinal cord with uniform density. Thus, the extent, organization, and even the existence of tree shrew motor cortex remained uncertain.

The purpose of the present study was to investigate more comprehensively the organization of motor and somatosensory cortex in Belanger's tree shrews (*Tupaia belangeri*), by using modern histochemical procedures to define architectonic subdivisions of frontoparietal cortex in combination with corticospinal tracing and detailed intracortical microstimulation (ICMS). This study is part of a larger effort to develop a model of the organization of sensory and motor cortical regions in tupaiids. Subsequent papers will present the results of the anatomical tracing experiments in these animals. The ultimate goal of these studies is to determine which features of tree shrew sensory and motor cortex are shared with primates and with mammals in general. Here we provide evidence that the frontoparietal cortex of tree shrews contains at least two distinct motor fields and five somatosensory fields. The motor cortical fields of tree shrews share a number of basic features with the primary and premotor cortical regions of primates, but evidence for the complex array of secondary motor fields identified in primates is lacking.

Abbreviations

3a	area 3a
3b	area 3b (primary somatosensory cortex)
Aud	auditory cortex
M1	primary motor cortex
M2	secondary motor cortex
PP	posterior parietal cortex
PV	parietal ventral area
S1	primary somatosensory cortex
S2	secondary somatosensory cortex
SC	somatosensory caudal area
TA	temporal anterior area
TD	temporal dorsal area
TI	temporal inferior area
TP	temporal posterior area
TPI	temporal posterior inferior area
V1	primary visual cortex
V2	secondary visual cortex

MATERIALS AND METHODS

The architecture, corticospinal connections, and movement topography of sensorimotor cortex were identified in a total of 11 Belanger's tree shrews of either sex (adults aged 1–2 years old, bred in our laboratory). Anatomical tracers injected into the spinal cord of three tree shrews revealed the pattern of corticospinal connections in six hemispheres. Intracortical microstimulation (ICMS) was used to identify the movement topography of frontoparietal cortex in eight tree shrews. In two cases, the anatomical injections were followed by ICMS mapping procedures. An additional two cases were used for more detailed architectonic analysis. The corticospinal and electrophysiological results were related to a number of histological features of cortex sectioned in the parasagittal, coronal, or tangential (flattened) plane (for summary see Table 1). All procedures were approved by the Vanderbilt University Animal Care and Use Committee and followed National Institutes of Health guidelines.

Spinal cord injections

For spinal cord injections, a surgical level of anesthesia was induced with a combination of ketamine hydrochloride (Ket; 90 mg/kg IP) and xylazine hydrochloride (Xyl; 10 mg/kg IM), and maintained by 1–2% Isofluorane. Ketoprofen (2.2 mg/kg) was administered prior to surgery to reduce inflammation and provide postoperative analgesia. An incision was made into the skin of the upper back, and the underlying muscles were retracted, exposing the spinal column. A laminectomy was performed over select lower cervical vertebrae (between C5 and C7) and a series of two to three injections of wheat germ agglutinin conjugated to horseradish peroxidase (WGA-HRP; 2% in distilled water) was placed across the exposed portions of the spinal cord on each side. Injections were spaced approximately 1.5 mm apart in the rostrocaudal dimension. Volumes of 0.4–0.5 μl were pressure injected at depths of 2 mm and 1.5 mm. Cervical injections were placed to maximize uptake between the C5 to T1 myelomeres. After injections, the exposed portions of spinal cord were covered with gel

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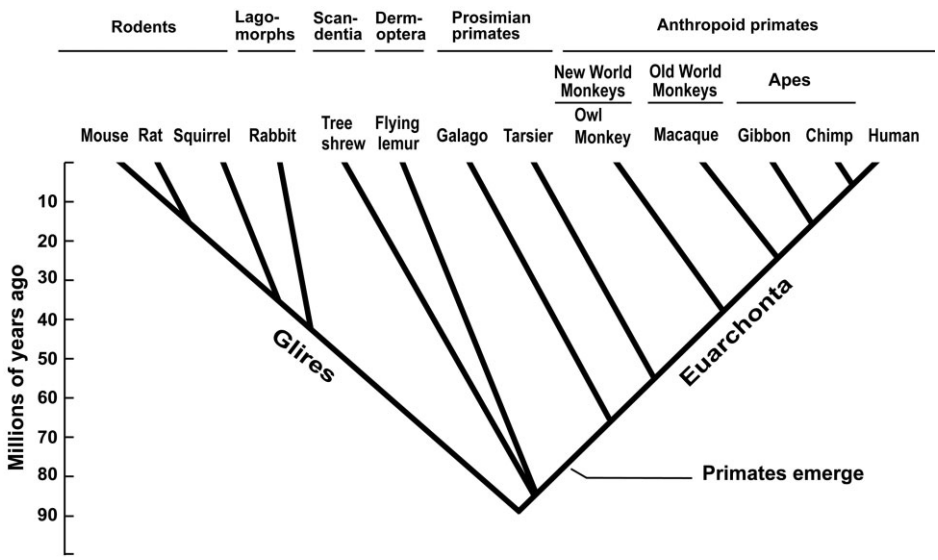


Fig. 2. Phylogeny of extant Euarchontoglire mammals. Superorders or clades are based on large-scale studies of the nuclear and mitochondrial genomes of extant mammal species (see, e.g., Murphy et al., 2001a; Waddell and Kishino, 2001). The Euarchontoglire clade combines the former Euarchonta clade of primates, tree shrews (Scandentia), and flying lemurs (Dermoptera) with the Glire clade (rodents and lagomorphs). The hierarchies and divergence ratios presented here are generally accepted, but absolute divergence times depend on calibration point. Divergence times presented here are based primarily on molecular estimates (Murphy et al., 2001a; Waddell and Kishino, 2001; Brunet et al., 2002; Huchon et al., 2002). © Michael S. Remple, with full rights granted to John Wiley & Sons, Inc.

TABLE 1. Summary of Experimental Cases[1]

Case No.	Spinal cord injection	Plane of section	Tracer	ICMS
102	—	Coronal	—	No
104	—	Coronal	—	No
310	C5-T1	Tangential (flat)	WGA-HRP	Yes
313	C5-T1	Tangential (flat)	WGA-HRP	No
320	C5-T1	Tangential (flat)	WGA-HRP	Yes
339	—	Parasagittal	—	Yes
345	—	Parasagittal	—	Yes
353	—	Parasagittal	—	Yes
411	—	Tangential (flat)	—	Yes
424	—	Parasagittal	—	Yes
428	—	Parasagittal	—	Yes

[1]ICMS, intracortical microstimulation; WGA-HRP, wheat germ agglutinin-horseradish peroxidase.

foam, and the overlying muscles and skin were sutured. Animals were carefully monitored for recovery from the anesthetic and then placed in a separate cage to recover. After 2–3 days, animals were prepared for ICMS (cases 310 and 320) or histology (case 313).

ICMS

Ket and Xyl were used to induce (30 mg Ket + 1.2 mg Xyl) and maintain (10 mg Ket + 0.2 mg Xyl) anesthesia for microstimulation procedures. A craniotomy was performed to expose the right frontal cortex, and the overlying dura was removed. A digital photograph was made of the cortical surface, and the brain was covered with warm silicon oil to prevent desiccation. A tungsten microelectrode (1 MΩ impedance) was used to make perpendicular penetrations into cortex to a depth of ~1500 μm, where the lowest-threshold movements were elicited (presumptive layer V). In five cases, the stimulus train consisted of 18 cathodal pulses of 0.2 msec duration (rate = 350 Hz) that were delivered every 1 second. In two cases, a shorter stimulus train consisting of 13 cathodal pulses was delivered (0.2 msec, 350 Hz, every 1 second), and in one case (411) both 18 and 13 cathodal pulses were delivered. Electrode penetrations were made with reference to a grid, at an interpenetration distance of 1,000 μm, vasculature permitting, and subsequently filled in to an interpenetration distance of 500 μm or less. This method was used to minimize excitability of adjacent regions of cortex due to current spread. Electrode penetrations were marked on a digitized photograph of the cortical surface, using the program Canvas 7 (Deneba, Miami, FL). The animal was carefully monitored for movement by two observers, and movements were classified into four categories. Oral movements included movements of the lips and tongue; facial movements included movements of the vibrissae, jaw, and eyelids; trunk movements included movements of the upper and lower trunk and neck; forelimb movements included movements of the digits, wrist, elbow, and shoulder; and hindlimb movements included movements of the toes, ankle, and knee. At each penetration, the minimal threshold required to elicit a movement was recorded, and sites where no movement was detected at 120 μA were recorded as unresponsive. A cut-off of 120 μA was set as a compromise between identifying secondary motor areas that require higher amplitude stimulation and limiting current spread in the cortex to less than 400 μm (Stoney et al., 1968). For case 411, movement thresholds were characterized for a shorter stimulus train (13 pulses), and then following a 2-minute refractory period the normal stimulus train was delivered (18 pulses) and the threshold movement was characterized again (ICMS cases summarized in Table 2). Electrolytic lesions (10 μA, 10 second) were made at select locations in the cortex to aid in subsequent reconstruction. Topographic maps of movement representation were gener-

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TABLE 2. Summary of Microstimulation Cases[1]

Case No.	ICMS parameters	Total penetrations	Penetrations (responsive/nonresponsive)				
			SC	3b	3a	M1	M2
310	N	90	9/6	22/5	10/2	8/3	6/8
320	N	141	14/7	24/12	12/3	19/7	15/3
339	N	196	12/6	41/9	19/2	44/7	13/15
345	N	123	3/7	10/12	8/3	16/5	7/7
353	N, L	132	3/2	29/2	16/1	24/2	15/14
411	S, N, L	106	16/2	25/4	10/3	12/2	6/3
424	S, L	114	3/4	25/2	10/5	19/2	8/4
428	S	86	—	3/0	8/1	28/3	17/9

[1]ICMS, intracortical microstimulation; N, normal; L, long; S, short. The total number of penetrations includes responsive and nonresponsive sites outside of the fields reported here.

ated from the pattern of electrode penetrations using the graphics program Illustrator CS (Adobe, San Jose, CA).

Histology

At the end of the experiment, tree shrews were given a lethal overdose of sodium pentobarbital (100 mg/kg IP) and perfused transcardially with phosphate-buffered saline (PBS), followed by 2% paraformaldehyde in PBS (pH 7.3) and then 2% paraformaldehyde and 10% sucrose in PBS. The brain was removed and the cortex was either separated from the brain stem and flattened between glass slides (five cases), or left intact for parasagittal or coronal sectioning. The cortex, thalamus, medulla, and cervical spinal cord were postfixed for 1–2 hours and then placed in 30% sucrose overnight for cryoprotection. All the pieces were cut frozen on a sliding microtome at a thickness ranging from 30 to 40 μm. In cases in which the cortex was flattened, it was sectioned in a plane tangential to the pial surface. In the remaining cases, the cortex was cut in a parasagittal or coronal plane (summarized in Table 1). In cases with spinal cord injections, alternate series of cortical sections were processed for myelin (Gallyas, 1979) and tetramethylbenzidine (TMB; to visualize WGA-HRP; Gibson et al., 1984). In the parasagittal cases, a series of cortical sections was processed for Nissl substance (with thionin), and alternating series were processed for myelin, cytochrome oxidase (CO; Wong-Riley, 1979), and acetylcholinesterase (AChE; Geneser-Jensen and Blackstad, 1971). In some cases, a series of sections was processed with the monoclonal antibody SMI-32 (Sternberger Monoclonals; lot No. 17, raised in mice) that recognizes a nonphosphorylated heavy neurofilament protein (Lee et al., 1988), which is expressed in a subset of cortical pyramidal neurons (Campbell and Morrison, 1989; Hof et al., 1996), and has been used to differentiate subdivisions of motor cortex in primates (Preuss et al., 1997; Geyer et al., 2000b). Sections from the spinal cord were sectioned in the horizontal plane and reacted for CO and TMB.

Anatomical reconstruction

Parasagittal cortical sections. Several histological procedures were used to delineate architectonic borders in parasagittally sectioned cortex, including Nissl, myelin, CO, AChE, and SMI-32. Nissl and SMI-32 were the most useful for defining the agranular, granular, and transitional zones in sensorimotor cortex (see Fig. 3), but the other histochemical procedures were useful for corroborating particular borders in ambiguous sections. Digital photomicrographs of sections were obtained by using a Nikon DXM1200 camera mounted to a Nikon E800 microscope. Digital images were adjusted for contrast using the levels command in Adobe Photoshop CS and then were imported into Adobe Illustrator CS. Micrographs were aligned with each other, and the pattern of blood vessels, and architectonic borders were identified and drawn for each section. The locations of borders were corroborated by viewing sections at high power with a microscope. The distances between architectonic borders, lesions, and rostral and caudal poles of cortex were measured from a common origin for each section using an Illustrator plug-in (CADtools; Hotdoor, Laguna Beach, CA). Measurements were made by following the curvature of cortex along layer V, and measurements were transferred into Cartesian coordinates for reconstruction in the tangential (or flattened) plane. ICMS maps were aligned with reconstructed maps by using the pattern of microlesions.

Coronal cortical sections. Architectonic borders were identified in coronally cut cortical sections processed for Nissl, CO, and AChE. The agranular, granular, and transitional zones in sensorimotor cortex were identified primarily from Nissl sections, but the other histochemical procedures were used to corroborate borders in ambiguous sections. Borders were corroborated as described above.

Flattened cortical sections. Flattened sections were processed for myelin, AChE, and in some cases CO. The modular organization of flattened cortex was best revealed by staining for myelin, which reveals several sensory areas of cortex. Outlines of sensory areas, large blood vessels, and microlesions (in ICMS cases) were made from photomicrographs in Illustrator CS. For tracer injection cases, WGA-HRP-labeled cells and large blood vessels were plotted under bright- and darkfield illumination on a microscope with a computer-controlled stage (Neurolucida, Colchester, VT). Labeled pyramidal cells were identified and plotted under darkfield illumination and

Fig. 3. Architecture from three adjacent cortical sections cut in the parasagittal plane in case 353 (composite photomicrograph). Sections are from frontoparietal cortex and have been processed for Nissl substance (**A**), SMI-32 (**B**), or myelin fibers (**C**). Agranular motor cortex is divided into rostral and caudal divisions (M2 and M1), based on differences in layer V, including pyramidal cell morphology and packing density. Primary somatosensory area 3b was identified by the density of granular layer IV, and area 3a formed a transitional region between M1 and 3b with well-developed layers IV and V. The somatosensory caudal area (SC) was identified based its less dense granular layer IV and less distinct layer Va band (see. Fig. 10 for reconstruction of this case). Panels include section numbers in upper left corner. Rostral is to the right. Scale bar = 500 μm. © Michael S. Remple, with full rights granted to John Wiley & Sons, Inc.

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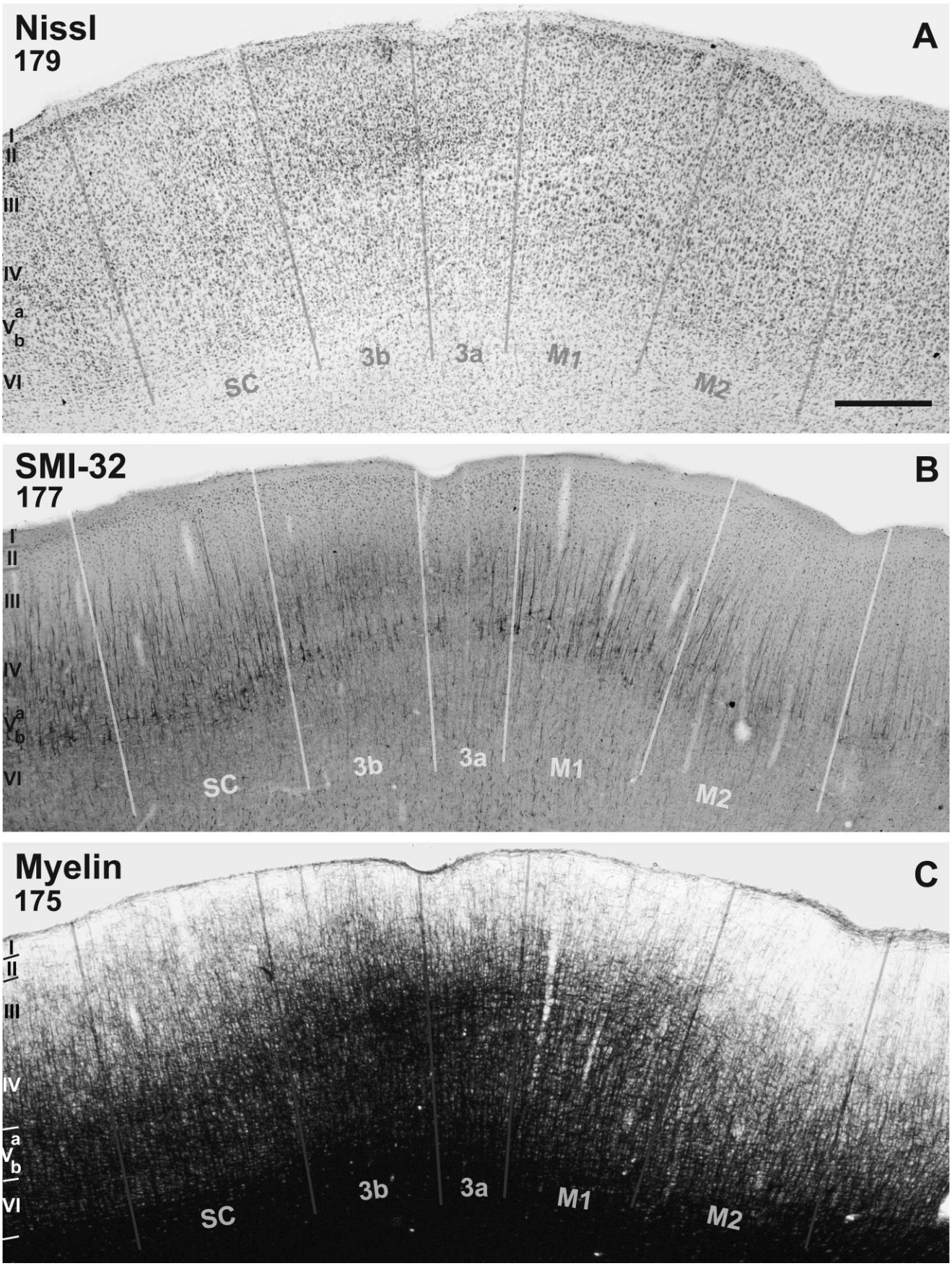


Figure 3.

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verified under brightfield illumination at higher magnification. Cells were recognized by their size, pyramidal morphology, and unique pattern of TMB labeling, which made their soma appear granular. TMB deposits were not classified as cells when they appeared uniformly dense, were irregularly shaped, or were notably different in size from adjacent unlabeled cells. Plotted sections were imported into Illustrator CS and aligned with photomicrographs of adjacent sections. Motor maps were aligned with myelin sections based on the pattern of microlesions. A solid line was drawn around primary somatosensory area 3b and a dotted line around the expected location of areas SC, 3a, M1, and M2, which were estimated in flat cortex based on their relative distance from area 3b.

Spinal cord. In injection cases, the cervical spinal cord was cut in a horizontal plane to best reveal the rostrocaudal and mediolateral zone of tracer uptake in individual sections. The zone of tracer spread was drawn from micrographs of TMB-processed sections and individual sections were superimposed on each other to determine the extent of tracer uptake. The zone of uptake was confirmed by examining sections at high power on a microscope under both darkfield and brightfield illumination.

Quantitative methods

The thresholds required to elicit movements at responsive sites were statistically analyzed in the program SPSS 13. Regions with fewer than three responsive sites were not included in threshold analysis. Thresholds were compared by using nonparametric statistical tests, because initial data exploration indicated that the data were not normally distributed (Shapiro-Wilk test of normality), and groups were unequal in size. In addition, many groups lacked homogeneity of variance (tested via Levene statistic). When the variance between groups was homogeneous, Mann-Whitney tests were used to compare two groups, and Kruskal-Wallis H-tests were used to test multiple groups. When the variance between groups was not homogeneous, Kolmogorov-Smirnov tests were used to compare two groups, and median tests were used to test multiple groups. Post hoc tests used either Mann Whitney (homogeneous variance) or Kolmogorov-Smirnov tests (nonhomogeneous variance), and a Bonferroni correction was applied so that all effects are reported at the 0.01 level of significance.

RESULTS

The results of histological reconstruction, intracortical microstimulation, and corticospinal tracing provide evidence for the subdivision of tree shrew sensorimotor cortex into seven distinct fields. We have identified an agranular region in frontal cortex that represents the motor cortex of tree shrews. This region can be further subdivided into primary and secondary motor cortex (M1 and M2, respectively) based on differences in architecture, movement thresholds, and the distribution of corticospinal neurons. We identified five fields in parietal cortex, including primary somatosensory cortex (area 3b, identified by Sur et al., 1980, as S1), a dysgranular region that is the likely homolog of area 3a in primates (Krubitzer et al., 2004), one region caudal to 3b that we have called the *somatosensory caudal area* (SC), and two fields in lateral cortex: the previously identified second somatosensory cortex (Sur et al., 1981a) and a more lateral area that may be the homolog of the parietal ventral area (PV) identified widely in mammals (Disbrow et al., 2000).

Cortical architecture

Parasagittal and coronal sections. The architectonic borders of motor cortex can be difficult to identify in tree shrews because of the relatively small size of Betz cells in these animals. Indeed, Sherwood et al. (2003) have shown that the ratios of Betz cell size to normal pyramidal neurons is smaller in tree shrews than in most mammals (Rivara et al., 2003). Therefore, we placed special emphasis on identifying motor regions in the forelimb region of tree shrew cortex, where layer V pyramidal neurons are largest and differences between somatosensory and motor regions are most pronounced. Architectonic borders were most reliably identified in sections cut in the parasagittal plane, because most borders were orthogonal to this plane. However, we also sectioned some cortices in the coronal plane to better reveal the differences in lateral cortex and some in the parallel (flat) plane, to relate the results of ICMS maps to the modular organization of somatosensory cortex.

Motor cortex. In Nissl-stained sections, motor cortex could be identified by using the classical criteria of a poorly developed granular layer IV and a well-developed pyramidal layer V. Differences in the appearance of layer V were used to subdivide motor cortex into primary motor cortex (M1) and a rostrally located secondary motor cortex (M2). In M1, layer V was thick with large, densely packed pyramidal neurons in deeper layer V (Vb) and a palely staining layer Va, whereas M2 had a thinner layer V with smaller more irregularly spaced pyramidal neurons and lacked an obvious pale layer Va (Figs. 3, 4). The rostral border of M2 could be identified in Nissl sections by the transition to smaller pyramidal cells, with a layer V thickness typical of nonmotor regions and a distinct granular layer IV. The SMI-32 antibody showed a high level of specificity for pyramidal cells in tree shrew cortex, similar to that described for primates (Campbell and Morrison, 1989; Preuss et al., 1997; Geyer et al., 2000b). In SMI-32 processed sections, layer V pyramidal neurons in M1 had large, densely immunoreactive cell bodies with thick apical dendrites that extended into layer III (Fig. 3), whereas layer V pyramidal cells in M2 had smaller, less densely labeled cells, with thinner apical dendrites that were more sparsely arranged than in M1. No immunoreactive cells were identified in layer III of either M1 or M2. The rostral border of M2 was identified by a transition to tall apical dendrites with few or no immunoreactive cell bodies. Pyramidal neurons in layer V stained darkly in CO preparations, but architectonic differences between M1 and M2 were not obvious. Differences between M1 and M2 were not apparent in AChE-processed sections, but, in favorable sections, the caudal border of M1 could be identified by a transition to lighter-staining layers III, IV and VI (Fig. 5).

Area 3a. A transitional region with a well-developed granular layer IV and pyramidal layer V was identified caudal to M1 and is likely a homolog of area 3a identified in primates (Krubitzer et al., 2004). Layer V pyramidal neurons were large in this region and transitioned from moderate to sparse packing density toward the caudal border of 3a, which was most evident in Nissl-processed sections (Figs. 3, 4). In SMI-32 processed sections, area 3a

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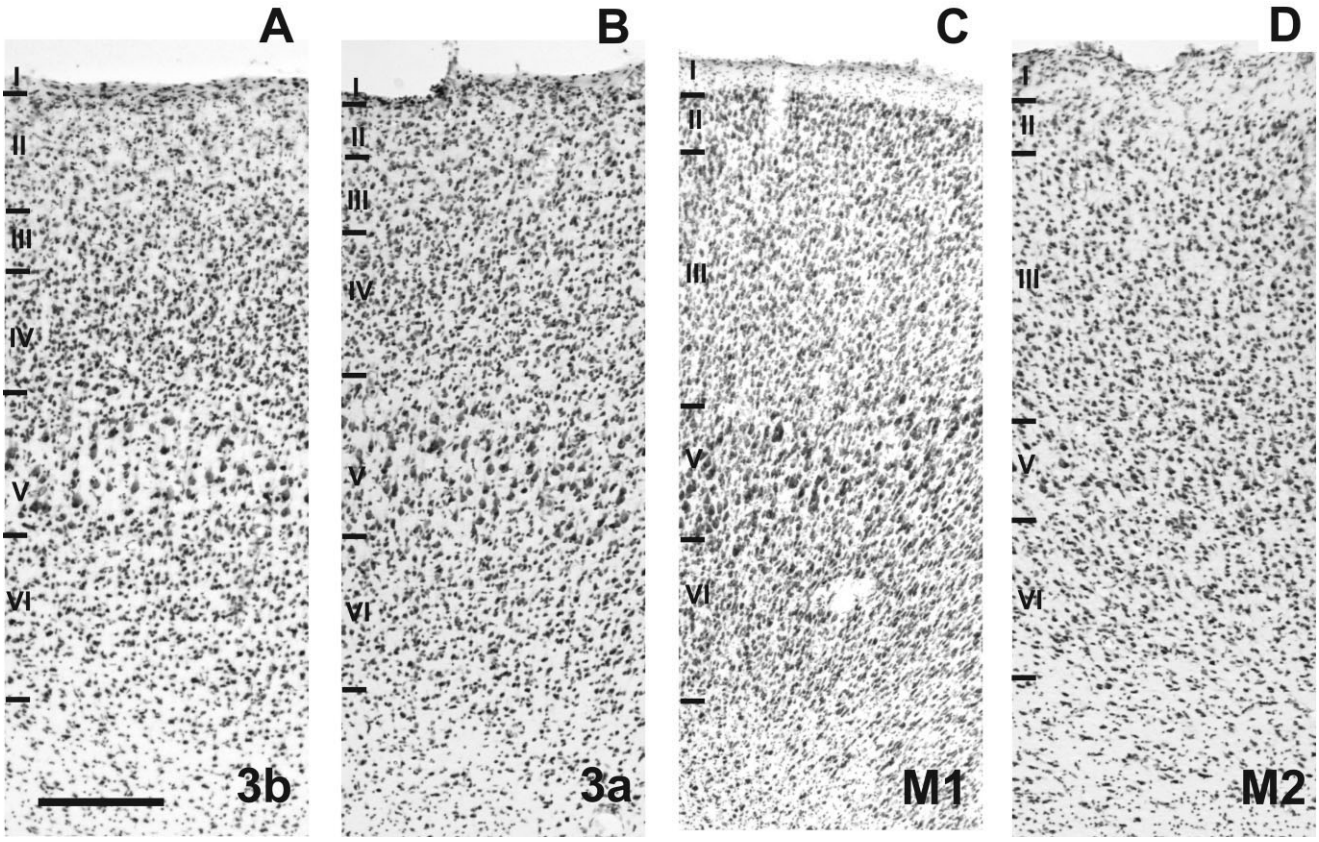


Fig. 4. Nissl cytoarchitecture from cortical sections cut in the coronal plane in two cases (102 and 104). **A:** Primary somatosensory area 3b has a dense granular layer IV and a pale layer V. **B:** Area 3a has a well-developed granular layer and layer V. **C:** Primary motor cortex (M1) is agranular, with large, densely packed layer V pyramidal neurons. **D:** Secondary motor cortex (M2) has large but sparsely arranged pyramidal cells. Case 104 is presented in A and B, and case 102 in C and D. Rostral is to the right. Scale bar = 250 μm. © Michael S. Remple, with full rights granted to John Wiley & Sons, Inc.

is differentiated from M1 by a lack of tall apical dendrites, small immunoreactive cells in layer IV, and a hazy pattern of labeling in the surrounding neuropil, with less densely arranged layer V pyramidal cells. A distinct, lightly stained band was evident in layer Va in both Nissl and SMI preparations (Fig. 3). In favorable myelin sections, area 3a could be differentiated from M1 by its darkly stained layer IV and from area 3b by its more lightly stained supergranular layers. In AChE-processed sections, area 3a could be differentiated from M1 by a dark layer V, surrounded by moderate staining bands above and below, corresponding to layers IV and VI (Fig. 5).

Area 3b. The primary somatosensory cortex (area 3b) was identified caudal to 3a and had a well-developed granular layer IV of uniform thickness. In Nissl-processed sections, layer V pyramidal cells were not as large as those in M1 and were more sparsely packed than in 3a, making the pale-staining layer Va more prominent (Figs. 3, 4). Area 3b appeared similar to area 3a in sections processed for SMI-32, but the haze in the granular layer IV was more pronounced, and cells in layer IV and their apical dendrites were more densely immunoreactive. In myelin-processed sections, 3b was identified as a band of particularly dense staining within layer IV, with an additional band of moderate staining in the more superficial supragranular layers (Fig. 3). A similar, but less distinctive, pattern of layer IV staining was identified in CO-processed sections (not illustrated). In sections processed for AChE, area 3b looked similar to area 3a, but layers III, IV, and VI were more lightly stained (Fig. 5). The caudal border of 3b could be identified in AChE sections by a transition to more uniform AChE staining and a darker layer VI.

Area SC. A region with a less-well-developed granular layer IV, which we have called the *somatosensory caudal area* (SC), was identified in the anterior parietal region of cortex just caudal to area 3b. In Nissl-processed sections, SC had a thick granular layer IV, with a moderate cell density and medium-sized layer V pyramidal cells that were sparsely distributed, giving layer V an overall pale appearance (Figs. 3, 4). In SMI-32 sections, the SC region had more densely immunoreactive layer V pyramidal neurons than in area 3b and a thinner layer Vb (Fig. 3).

Lateral cortex. The second somatosensory and parietal ventral areas (S2 and PV, respectively) were identified in lateral portions of sections cut in the coronal plane. The second somatosensory area S2 was located lateral to the face region of 3b and could be differentiated from 3b based on its thinner, semipale layer V that had smaller pyramidal neurons, and thinner, less-well-developed granule

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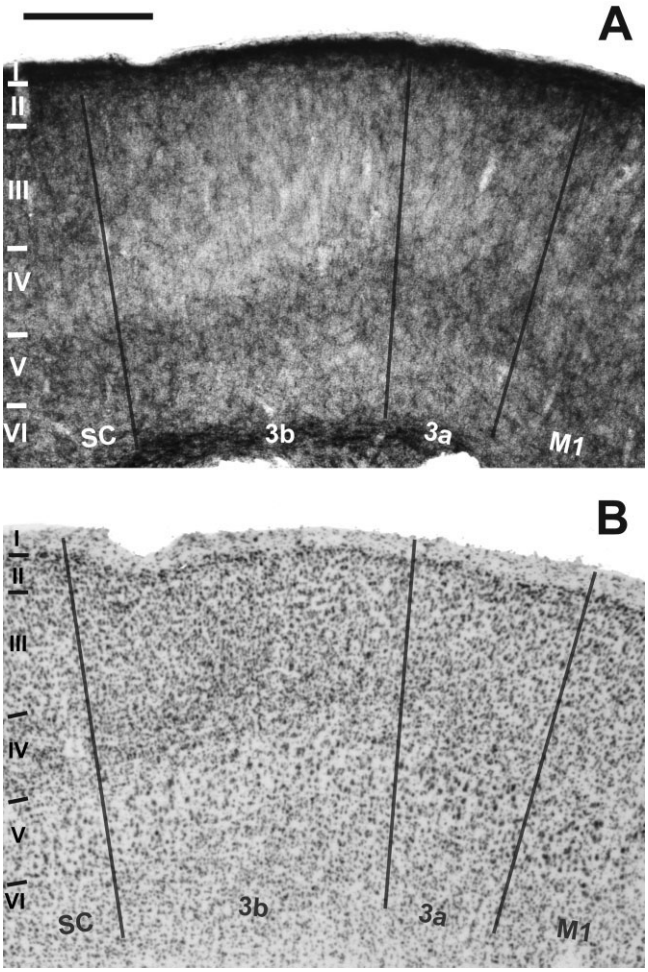


Fig. 5. Architecture from two adjacent cortical sections cut in the parasagittal plane in case 428. Section are processed for acetylcholinesterase (AChE; **A**) and Nissl substance (**B**). Note the distinctive pattern of AChE staining in area 3b, where layers IV and VI stain lightly and layer V stains darkly. A similar but less distinctive pattern of staining is seen in area 3a. The adjacent Nissl-stained section is provided for comparison. No clear distinctions were noted in AChE sections for areas M1 and M2. Rostral is to the right. Scale bar = 500 μm. © Michael S. Remple, with full rights granted to John Wiley & Sons, Inc.

layer IV (Fig. 6). Laterally to S2, the parietal ventral (PV) region could be identified based on the appearance of layer V, which was thinner and more densely packed and lacked the pale band in layer Va.

Flattened cortical sections. Although no histochemical correlates of motor cortex were identified in flattened cortex, this preparation was valuable for identifying a number of sensory subdivisions of cortex (Fig. 7A,B). In myelin-stained, flattened sections, 3b appeared as a myelin-dense band starting at the medial wall of cortex and traveling rostrolaterally to the edge of neocortex, with myelin-sparse septal regions dividing the face from the hand region (Cusick et al., 1985), the hand from the foot, and the face from the mouth in 3b (Sur et al., 1980, 1981b). A series of myelin-dense modules, separated by myelin-sparse septal regions (arrowheads in Fig. 7D), was identified lateral to the face region of 3b that corresponded to the expected location of the hindlimb, forelimb, and oral representations of S2 and possibly PV (Fig. 7B,D). We have labeled these modules *OM* (oral module), *FM* (forelimb module), and *HM* (hindlimb module). Similar modules have been identified in the lateral cortex of rodents (Remple et al., 2003; Henry et al., 2006). Myelin staining also revealed a number of other sensory regions, including V1,V2, TI, and auditory cortex (Kaas et al., 1972; Lyon et al., 1998). The myelin-dense sensory regions were most clearly identified in the middle to deeper layers of cortex, which correspond to layers IV and V in parasagittal sections (Fig. 3). A similar but less robust pattern of staining was identified in flat sections processed for CO (not illustrated). In AChE-processed sections, area 3b appeared as a lightly stained strip that was identified in both superficial and deep sections and formed a pattern inverse to that identified in myelin processed sections (not illustrated). The pattern of AChE staining was a less reliable indicator of sensory regions than myelin, however, as the AChE-light pattern was found in a more limited range of sections and was often incomplete in single sections.

Corticospinal neurons

The fast transporting retrograde tracer WGA-HRP was injected into the lower cervical spinal cord of three tree shrews to retrogradely label corticospinal cells in the forelimb region of sensorimotor cortex. We sectioned hemispheres in the tangential plane to allow for the most accurate reconstruction of corticospinal cell topography. The distribution of corticospinal cells was related to the location of primary somatosensory cortex and estimates of the adjacent motor and somatosensory fields. Corticospinal cells were identified in multiple somatosensory and motor areas but were most densely distributed in 3b and the area just rostral to it (i.e., the expected location of M1 and 3a). The densest concentration of corticospinal cells was found in the central forelimb region of both M1 and 3a, but a less-dense focus of cells was identified at the ventral edge of the forelimb region, where both forelimb and orofacial movements were elicited (Fig. 8; see ICMS section). An interesting pattern of corticospinal labeling was identified in the hand region of 3b, which had a relatively sparse focus of labeled cells in the center, surrounded by a region of dense cell labeling. The region of sparse labeling in 3b corresponded to the location of the digits described by Sur et al. (1980), whereas the dense corticospinal labeling in the surrounding region corresponded to the location of arm, wrist, and the pads of the hand (Sur et al., 1980). A more sparse distribution of corticospinal cells was often found in the regions surrounding the hand and foot regions in 3b, where the trunk, arm, and leg are represented (Sur et al., 1980, 1981b). Limited numbers of corticospinal neurons were identified in the expected location of M2 following lower cervical injections. A patchy distribution of corticospinal neurons was consistently identified caudal to the 3b forelimb region in the expected location of SC. Corticospinal cells were not always identified in the S2/PV region of lateral cortex, but a few scattered cells were identified in this region in two hemispheres (320L and 320R; Fig. 8).

Microstimulation maps

Intracortical microstimulation (ICMS) was used to determine the organization and extent of motor cortical ar-

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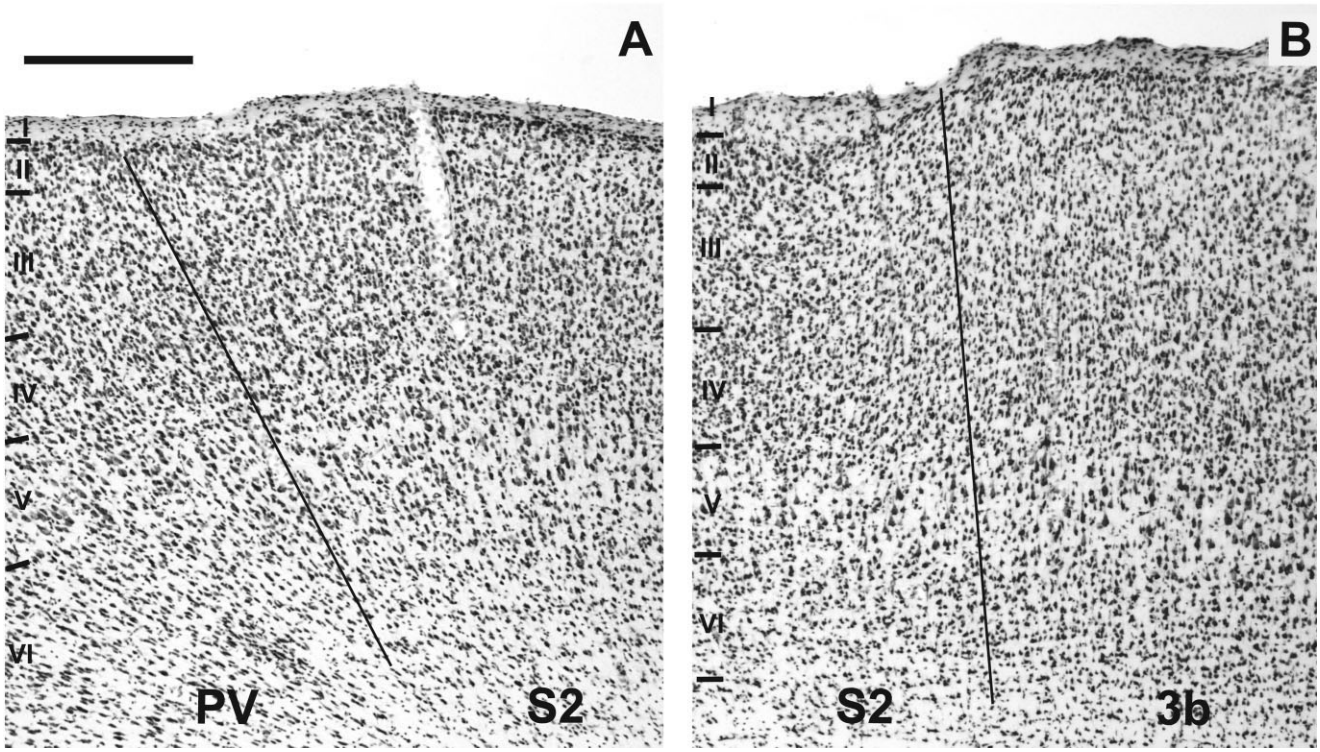


Fig. 6. Cytoarchitecture of lateral cortex, in a section cut in the coronal plane and processed for Nissl substance (case 104). **A:** The second somatosensory area (S2) is more stratified overall and has a sparser packing density of layer V pyramidal neurons than the more lateral parietal ventral area (PV). Layer V pyramidal neurons in PV are smaller and less densely packed than in the more medial second somatosensory area (S2). **B:** In area 3b, granular layer IV is thicker and more dense, and layer V pyramidal neurons are larger and more densely packed than in S2. Rostral is to the right. Scale bar = 500 μm.
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eas in tree shrews. In total, 988 penetrations were made in eight hemispheres for the present study, with individual cases ranging from 86 to 196 penetrations (Table 2). In five cases, we used ICMS parameters typically used in primates (see, e.g., Stepniewska et al., 1993) so that we could more directly compare our results with those of primate ICMS studies. These "normal" stimuli consisted of 54-msec trains of 18 cathodal pulses, delivered every 2 seconds. The effects of shorter stimulus trains (39 msec, 13 pulses) were also explored in three tree shrews, because these parameters are more typical for ICMS studies in smaller-brained animals (see, e.g., Remple et al., 2001), but the evoked movements and movement thresholds did not differ significantly from those found in normal stimuli cases (Mann-Whitney test). Thus, the results of normal- and shorter-stimuli groups were combined. The effects of longer stimulus trains (500 msec or 150 pulses) were explored at select locations in three cases (see Table 1), because these parameters have recently been shown to produce maps of complex multifaceted movements in primate motor (Graziano et al., 2002) and intraparietal cortex (Cooke et al., 2003; Stepniewska et al., 2005). In the present study, however, movements generated by long stimulus trains were not qualitatively different from those produced by normal stimuli at the same locations, so the results of long trains were not explored in detail. The injection of anatomical tracers into the cervical spinal cords of two cases (310 and 320) did not appear to impact the excitability of motor cortex, insofar as thresholds for forelimb movements in injected cases were not significantly different from those of all noninjected cases or stimulus-matched cases (Mann-Whitney test).

General features of ICMS maps. Movements were elicited from a wedge-shaped region of frontoparietal cortex that extended 4–5 mm mediolaterally and 3–5 mm rostrocaudally. This excitable region was 18 mm^2, on average, and typically straddled two branches of the large cortical vein in frontal cortex that connects to the superior sagittal sinus (Poonkhum et al., 2000). A cortical indentation was identified under the rostral branch of this vein, corresponding to the location of the orbital furrow identified by Lende (1970). The excitable region was found primarily on the dorsal surface of cortex, but the rostral most region extended partway into lateral cortex. No movements were identified in the medial wall of cortex.

Elicited movements were broadly somatotopic in excitable cortex, with a medial to lateral progression of hindlimb, forelimb, face, and mouth (Figs. 9–11). No precise somatotopy was found within these regions, however, as the movement topography formed the fractured mosaic typical for ICMS mapping studies of sensorimotor cortex in other mammals (e.g., Gould et al., 1986). That is, similar movements were scattered widely across somatotopic divisions, and dissimilar movements were often located adjacent to each other. Most identified movements were "simple," occurring around a single joint, but occasional

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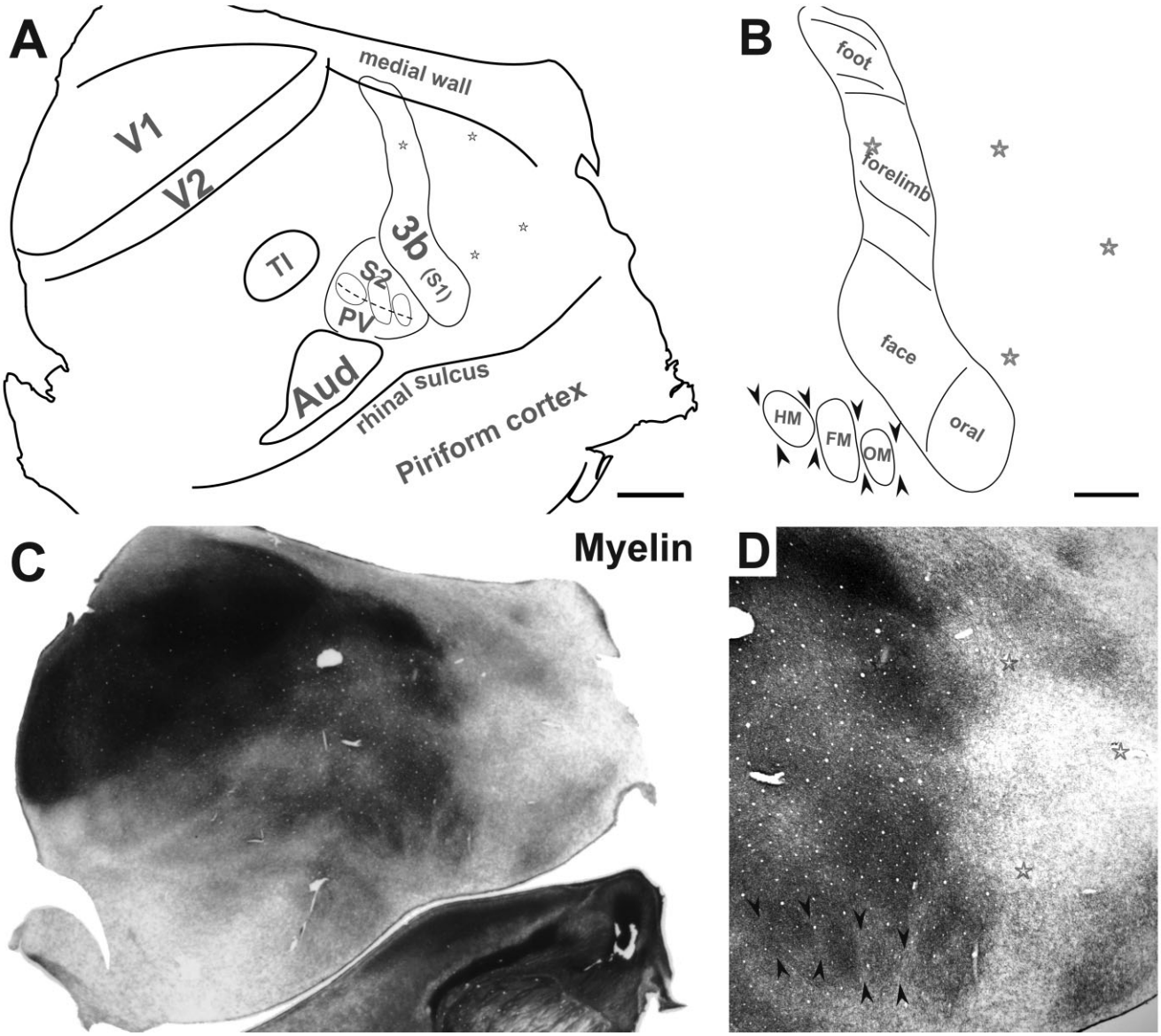


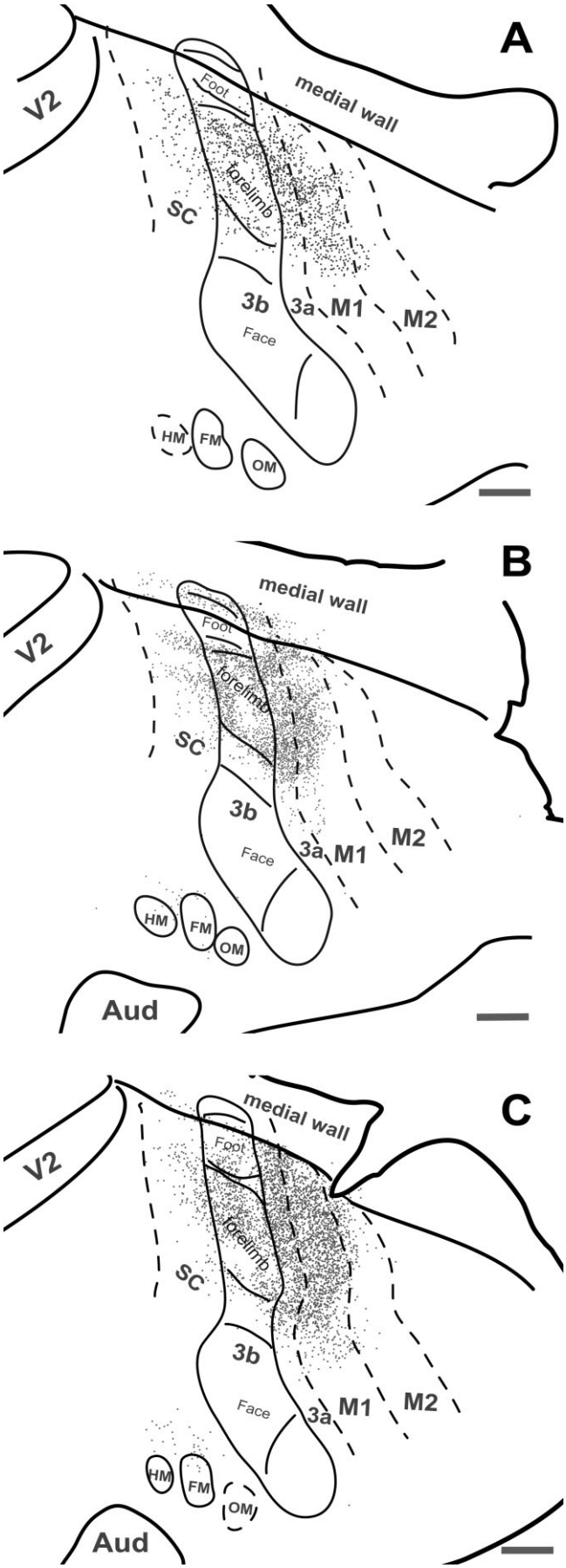
Fig. 7. The modular organization of sensory areas in flattened cortex in case 320. **A:** Summary of sensory modules revealed by myelin processing. **B:** Somatotopy of primary somatosensory region 3b. **C:** Sensory areas can be visualized as myelin-dense modules in the middle layers of flattened cortex. **D:** A close-up of area 3b reveals myelin-dense regions divided by myelin-light septal regions that define borders of somatotopic divisions of area 3b and modules in lateral cortex. Arrowheads in B and D indicate septal regions separating the hindlimb, forelimb, and oral modules in S2/PV (HM, FM, OM, respectively). Stars indicate microlesion sites used for alignment. Aud, auditory cortex; S2, secondary somatosensory cortex; TI, temporal inferior region; V1, primary visual cortex; V2, secondary visual cortex. Rostral is toward the right and medial is toward the top. Scale bars = 2 mm in A (applies to A,C); 1 mm in B (applies to B,D). © Michael S. Remple, with full rights granted to John Wiley & Sons, Inc.

"complex," multijoint movements, and dual movements of two separate body parts were identified. At threshold levels, most elicited movements were contralateral to the stimulated hemisphere, but bilateral movements were frequently identified at midline structures of the head, such as the vibrissae, jaw, and tongue. Many penetrations yielded complex, dual or bilateral responses at suprathreshold levels. The median threshold required to elicit movements differed significantly between regions [$H(5) = 39.6$, $P < 0.001$], and the results of post hoc tests are reported below (a Bonferroni correction was applied so that regional effects are reported at a 0.01 level of significance).

Motor cortex

M1. The primary motor cortex formed a narrow medial to lateral strip that was approximately 1 mm wide, with the body represented mediolaterally and the orofacial region represented rostrolaterally. Although there were no obvious morphological features to help locate M1 from the brain surface, its location could be approximated with reference to a distinctive skull suture over occipital cortex.

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The medial portion of M1 was typically identified 1 mm lateral and 5–6 mm anterior to this suture, and the orofacial region was 5 mm lateral and 3–4 mm anterior to it. In general, movement thresholds in M1 were lower than any other cortical region; the median threshold was 24 μA across cases, and 67% of M1 thresholds were below 40 μA. The median threshold for M1 was significantly lower than that of the adjacent M2 (36 μA; K-S $z = 1.98$, $P < 0.001$) but was not significantly different from the caudally located 3a region (27.5 μA). These statistical results are consistent with our observations during ICMS mapping, insofar as the border between M1 and M2 could often be identified by a clear transition to higher thresholds, but no transition was noted for the caudal border between M1 and 3a (e.g., Fig. 9B). Thus, subsequent histological reconstruction was required to identify M1 conclusively.

The medial portion of M1 was primarily devoted to the forelimb representation, with a few scattered trunk and neck movements. No hindlimb movements were identified in M1 in any case. Most forelimb movements were elicited at moderately low thresholds, with a median value of 26.5 μA, and 68% of these movement fell within a range of 3–40 μA. The forelimb representation was almost exclusively devoted to proximal movements of the elbow and shoulder, with only 6% of forelimb sites yielding distal movements. Because of the limited number of distal responses, statistical comparisons were not made between the proximal and distal movement categories. When distal movements were found, they were usually simple flexions of the wrist or all of the digits, although occasional supination movements of the wrist were observed.

A large orofacial representation was identified lateral to the forelimb region in M1, where tongue, lip, jaw, and whisker (i.e., vibrissae) movements were elicited. Oral movements were the most prevalent in this region, particularly movements of the tongue, which occurred in isolation or in combination with jaw movements. Thresholds for oral movements were very low, with a median threshold of 11 μA across cases, and 30% of the movements were elicited at thresholds between 1–6 μA. Facial movements were less common and were mostly movements of the vibrissae on the snout and chin, which were usually found on the dorsal portion of the orofacial region. Eye-blink movements were not identified in M1 in any cases.

M2. The M2 region formed a tapered strip just anterior to M1, which was narrower and less extensive in the medial dimension. The median threshold for M2 movements was 36 μA, which as already noted was significantly higher than the median threshold in M1. The rostral border of *lateral* M2 could be identified by a transition from orofacial movements to unresponsive sites, which agreed with borders defined by histological reconstruction. However, the rostral border of *medial* M2 was not always

Fig. 8. Representative pattern of corticospinal cells labeled after bilateral injections of WGA-HRP into the cervical spinal cord (C5–T1) of cases 310 (right hemisphere, **A**) and 320 (right and left hemispheres, **B** and **C**, respectively). Solid lines indicate boundaries of sensory regions identified from alternate sections; dashed lines indicate borders estimated from parasagittal cases. Note the dense labeling in the caudal division of primary motor cortex (M1) and area 3a and the patchy labeling in area 3b. Rostral is toward the right, and medial is toward the top. Scale bar = 1 mm. © Michael S. Remple, with full rights granted to John Wiley & Sons, Inc.

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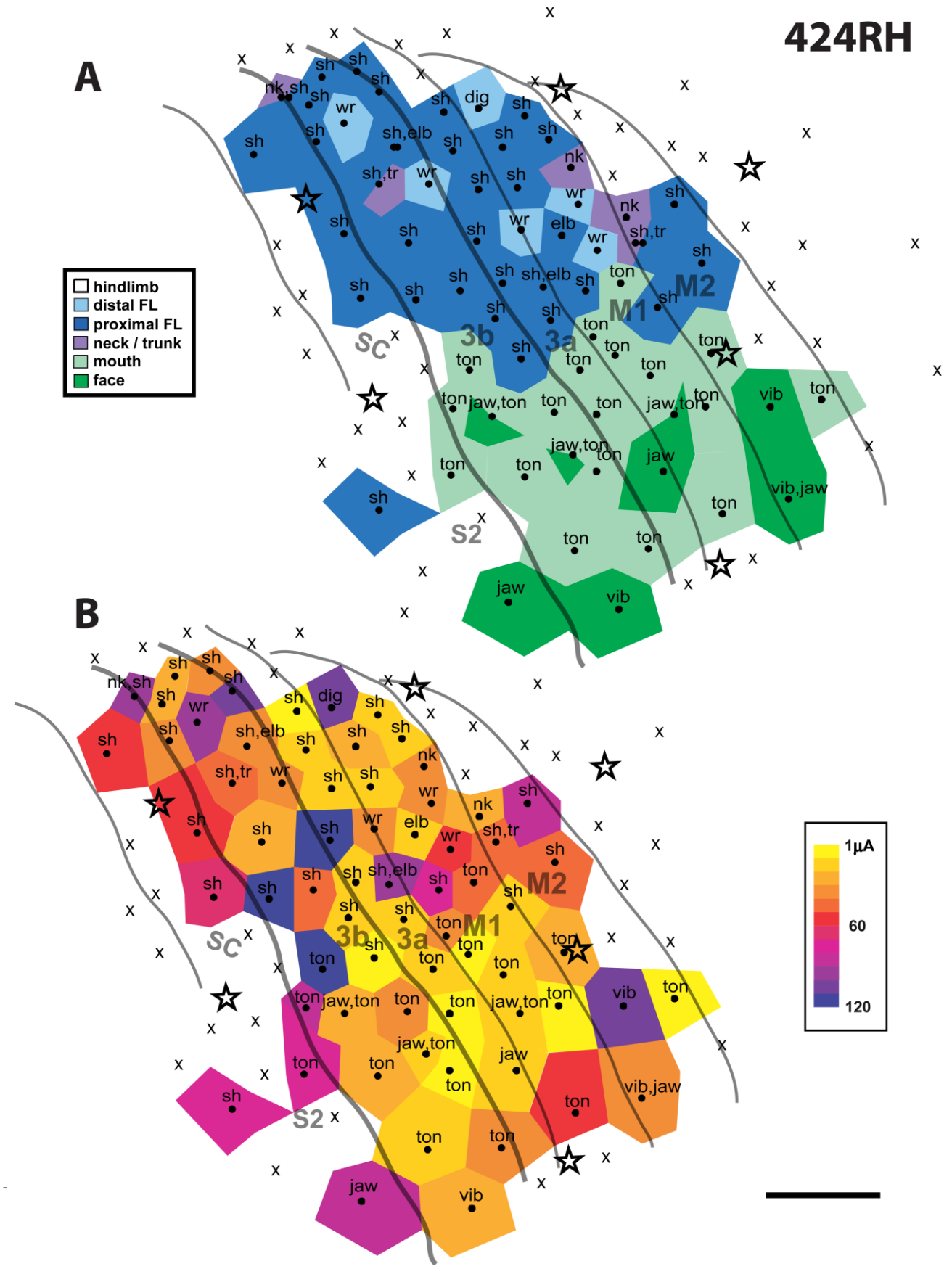


Figure 9.

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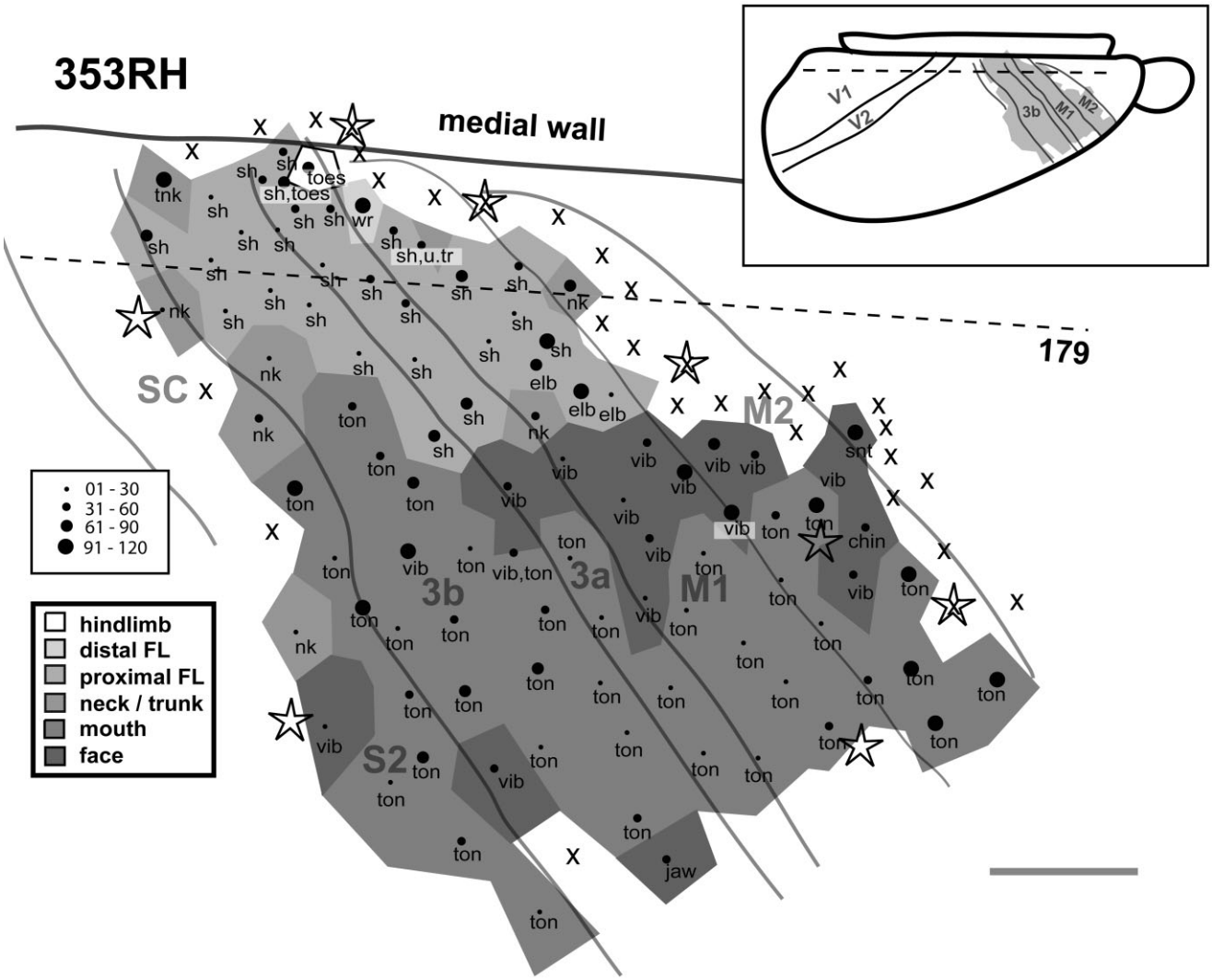


Fig. 10. Movement topography derived from intracortical microstimulation (ICMS) in the right hemisphere of case 353. Solid lines indicate location of architectonic borders reconstructed from sections cut in the parasagittal plane. The dashed line indicates the plane of section 179 that is illustrated in Figure 3. Circle size indicates the minimal threshold range required to elicit movements. Abbreviations are the same as those used for Figure 9. Rostral is toward the right, and medial is toward the top. **Inset:** Location of motor map in relation to reconstructed dorsal view of the brain. Scale bar = 1 mm. © Michael S. Remple, with full rights granted to John Wiley & Sons, Inc.

apparent during ICMS mapping, as unresponsive sites were often found within this region. It is likely that movements would have been elicited from some of these unresponsive sites if the cut-off threshold was higher than 120 μA; however, it was decided that this threshold was necessary to limit the effects of current spread (Stoney et al., 1968) and minimize the potential for tissue damage. The movements identified in the medial portion of M2 were similar to those found in medial M1, with forelimb, trunk, and neck representations and no hindlimb representation. Thresholds needed to elicit forelimb movements in M2 were significantly higher than in M1 (K-S $z = 1.98$, $P < 0.001$), with a median threshold of 45 μA, with only 30% at or below 40 μA. As in medial M1, forelimb movements were mostly proximal in M2, and only a few sites were classified as distal (9% of forelimb responses). A greater proportion of medial M2 was devoted to movements of the trunk and neck, and these movements were elicited at higher thresholds (90 μA median) and were often found at the rostral edge of the map. The lateral orofacial representation of M2 had an organization similar to that of M1.

Fig. 9. Movement topography and threshold maps derived from intracortical microstimulation (ICMS) in the right hemisphere of case 424. Maps are related to architectonic borders reconstructed from sections cut in the parasagittal plane. The movement topography is indicated in **A**, and the thresholds required to elicit those movements are indicated in **B**. Circles indicate electrode penetrations; crosses indicate unresponsive penetrations (120 μA); stars indicate the location of microlesions used to align maps to histology. sh, Shoulder; dig, digit; u. tr, upper trunk; wr, wrist; vib, vibrissae; ton, tongue. Rostral is toward the right and medial is toward the top. Scale bar = 1 mm. © Michael S. Remple, with full rights granted to John Wiley & Sons, Inc.

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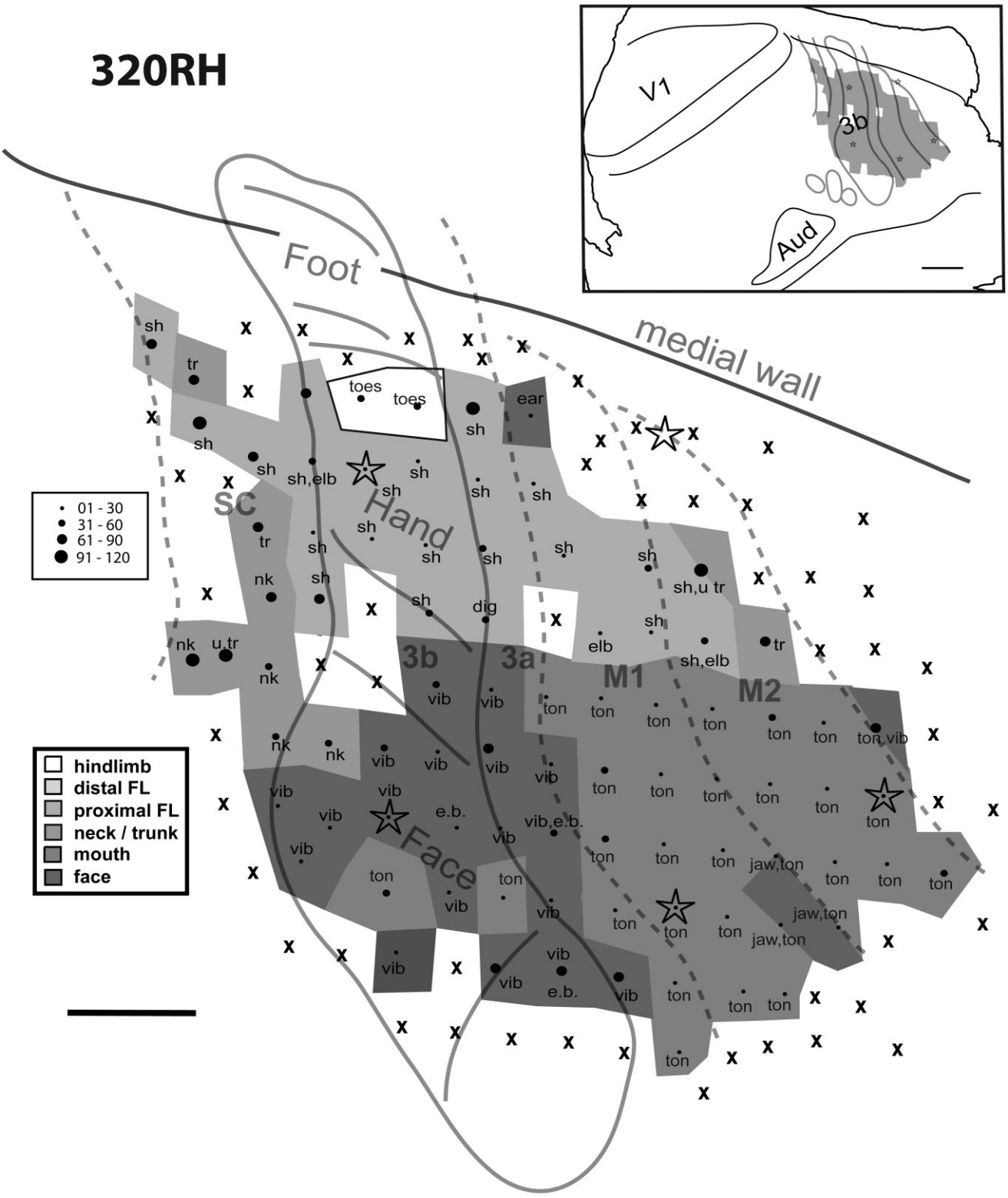


Fig. 11. Movement topography derived from intracortical microstimulation (ICMS) in case 320. Solid lines indicate location of borders of primary somatosensory cortex 3b in flattened cortex that were identified from myelin-processed sections (see Fig. 7). **Inset:** Location of motor map in relation to flatted cortex. Details and abbreviations are the same as those used for Figures 9 and 10. Rostral is toward the right, and medial is toward the top. Scale bar = 1 mm. © Michael S. Remple, with full rights granted to John Wiley & Sons, Inc.

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Movements of the mouth were highly represented in this region, with many penetrations eliciting tongue movements in isolation or in combination with facial or forelimb movements.

Anterior somatosensory cortex. The somatosensory areas 3a and 3b were located just caudal to M1 in anterior parietal cortex and were identified primarily by architectonic features in sections cut in the parasagittal and tangential plane. The primary somatosensory area (3b) formed a narrow strip that was approximately 1 mm wide at the foot and forelimb representations and 1.5 mm wide at the face and mouth representations. The forelimb and face representations in 3b are located on the dorsal surface of cortex, whereas the oral representation extends into lateral cortex, and the foot representation extends into the medial wall of cortex (Sur et al., 1980, 1981b). Area 3a formed a narrow strip between M1 and 3b that was approximately 0.5 mm wide. The somatosensory organization of 3a was not evident from histological analysis, but movements evoked by ICMS indicated a topographic organization with the body represented medially and the face represented laterally. No obvious transition was identified between areas 3a and 3b, and their respective median thresholds of 27.5 μA and 30 μA were not significantly different from each other.

The movement topography of lateral 3a and 3b was like that of lateral M1, with the exception that hindlimb movements could be elicited within the 3b foot representation and the region just anterior to it in 3a (see Figs. 10, 11). Hindlimb movements were identified in all but one case and tended to be flexions of the toe or ankle, although hip movements were found in case 320 (not illustrated). In most cases, the hindlimb representation was at the most medial portion of excitable cortex (e.g., Figs. 10, 11), but, in one case (411), the hindlimb representation was bordered by a more medial representation of the forelimb (not illustrated). The median hindlimb thresholds were 40 μA in area 3a and 45 μA in area 3b. The forelimb representations of 3a and 3b were organized similarly to each other and to M1, in that proximal forelimb movements predominated, with no distal movements identified in 3a across cases and only 6% identified in 3b. Few trunk and neck movements were elicited in either 3a or 3b. Orofacial movements were identified in lateral 3a and 3b regions and were similar to those identified in lateral M1, but, in two cases (310, 411), area 3b had a greater representation of the face than 3a or the motor regions (e.g., Fig. 11). Thresholds were surprisingly low for movements of the mouth in this region, with a median values of 25 μA in 3a and 31.5 μA in 3b.

Caudal somatosensory cortex. The caudal somatosensory area was located just caudal to 3b in parietal cortex and formed a strip approximately 1.4 mm wide (Fig. 3). This region was identified primarily from histological reconstruction, but in some cases differences in movement thresholds were apparent during ICMS mapping between SC and 3b. The median threshold for SC was 54 μA, which was significantly higher than that of 3b (K-S $z = 2.688$, $P < 0.001$). Movements were limited primarily to those of the trunk, neck, and proximal forelimb, with occasional orofacial movements. Movements were primarily identified in the rostral portion of SC, as most penetrations in the caudal portion of SC were nonresponsive.

DISCUSSION

The goal of the present study was to determine the number and organization of the motor and somatosensory fields in the Belanger's tree shrew (*Tupaia belangeri*). We used intracortical microstimulation (ICMS) to map the representation of movement in frontoparietal cortex and placed injections of anatomical tracers in the lower cervical spinal cord to determine the corticospinal connections of the forelimb region. The ICMS derived maps and the distribution of corticospinal cells were related to architectonic borders identified in histological sections processed by using a variety of histochemical procedures. The results provide evidence for the subdivision of tree shrew sensorimotor cortex into seven distinct fields, which are illustrated in Figure 12. Four of these fields are newly identified, including primary and secondary motor cortex (M1 and M2), the parietal ventral area (PV), and the caudal somatosensory area (SC).

Motor cortex in tree shrews

Several investigators have attempted to identify the motor cortex of tree shrews by using cytoarchitecture, corticospinal tracing, or electrophysiology, but the exact location and organization of this region has remained uncertain. Le Gros Clark (1924) identified a "Betz cell" region of frontal cortex that had large layer V pyramidal neurons, which he suggested was likely the motor region of tree shrew cortex, comparable to area 4 identified by Brodmann as the primary motor region of humans (Brodmann, 1909; Le Gros Clark, 1924, 1934). Le Gros Clark (1934) also described a granular region in parietal cortex that he suggested was likely a somatosensory zone. A subsequent analysis of the myeloarchitecture of this region by Zilles (1978) provided further support for separate motor and somatosensory fields in tree shrew cortex. However, this distinction was not apparent when surface macroelectrodes were used to stimulate the cortex in tree shrews, as Lende (1970) identified a broad zone where movements could be evoked at low thresholds, and this region was coextensive with much of the somatosensory cortex (identified by evoked potential recordings). Lende (1970) concluded that the motor and somatosensory regions of tree shrews were "an indivisible region of true overlap," similar to the highly overlapping sensorimotor amalgam that he proposed for opossums and hedgehogs. This sensory-motor overlap was not related to architecture but apparently included the M1, 3a, and 3b regions identified in the present study. The distinction between the motor and the somatosensory regions was also not apparent from the distribution of the corticospinal neurons described by Nudo and Masterton (1990), who included tree shrews in their landmark comparative study. They described a single, dense focus of corticospinal neurons in the frontal parietal cortex (corresponding to our M1, 3a, and 3b regions) surrounded by thin cell-sparse regions (our M2 and SC). The existence of a large, uniform focus of corticospinal neurons is consistent with earlier results of Jane et al. (1969), who found that lesions to either frontal or parietal cortex caused relatively equal patterns of degeneration of corticospinal terminations in the spinal cord.

The results of the present study provide clear evidence for the presence of distinctive motor and somatosensory fields in the frontoparietal cortex of tree shrews. More-

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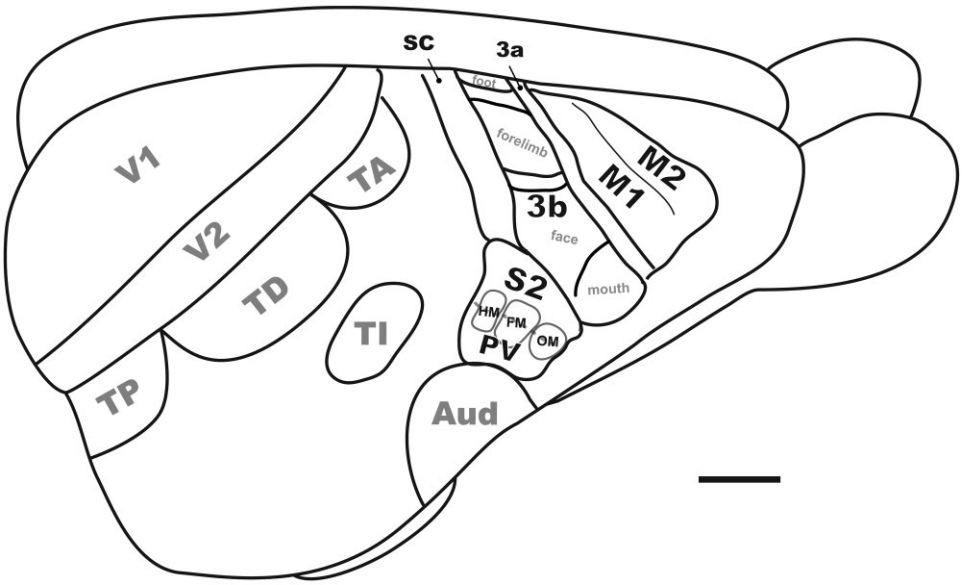


Fig. 12. Proposed model for the sensorimotor organization of tree shrew cortex. Rostral is toward the right, and medial is toward the top. Scale bar = 2 mm. © Michael S. Remple, with full rights granted to John Wiley & Sons, Inc.

over, the combined use of anatomical and electrophysiological criteria indicates that this region contains multiple cortical fields. Our histological analysis indicates that tree shrews have a well-developed motor region with many of the hallmark architectonic features identified in the motor cortex of other species. We identified within this region a caudal field that is the likely homolog of the primary motor cortex (M1)—an area that is identified widely in placental mammals—and a rostral field that we identify as secondary motor cortex (M2). The M1 region has a layer V that is tightly packed with pyramidal neurons that are densely immunoreactive, whereas the M2 region has large, but less tightly packed pyramidal neurons that are less immunoreactive. Architectonic differences between M1 and M2 were most pronounced in sections cut in the parasagittal plane, but they could also be identified in coronal sections. The extent of our motor cortex corresponds closely to the praecentralis 1 region of Zilles (1978), although he did not subdivide this region into two fields and defined the lateral extent of this region more narrowly. The distribution of corticospinal neurons appears markedly different between M1 and M2, with M1 sending dense projections to the lower cervical spinal cord and M2 sending only sparse projections to the cervical cord. It is possible that the M2 region may send more dense projections to higher cervical levels, but this conclusion is not supported by the findings of Nudo and Masterton (1990). They placed HRP crystals at the upper cervical (C1–C2) levels of the spinal cord of tree shrews and described a similarly sparse pattern of labeling in frontal cortex (in the expected location of M2). The densities of corticospinal neurons are relatively similar among M1, 3a, and 3b, but 3b appears to have a more patchy distribution of corticospinal neurons.

Electrophysiological results from this study indicate that thresholds are only partially useful for identifying differences between cortical regions in tree shrews. The clearest transition in movement thresholds occurs between the M1 and M2 fields, where M2 thresholds are significantly higher, particularly in the forelimb/body region. The transition between M1 and the adjacent somatosensory fields is less clear, and we did not detect statistical differences in thresholds among M1, 3a, or 3b. Given the similar density of corticospinal cells across these areas (at least for the forelimb region), this result is not surprising. In a broad sense, the movement topography in tree shrew motor cortex conforms to the pattern identified in most mammals. Movements are broadly topographic, with the body represented medially, and the face represented laterally, but individual movements within somatotopic divisions exhibit the fractured mosaic organization typically described in ICMS studies. The movements elicited in M1 and M2 are generally similar, but a transition to more axial movements of the body is apparent in the medial portion of M2. Several topographic features of tree shrew motor cortex are worth discussing further, including 1) the lack of an oculomotor and hindlimb representation, 2) the limited representation of the distal forelimb, and 3) the relatively large representation of the mouth. The significance of these findings is discussed in the following section, along with methodological issues that might have contributed to these findings.

Do tree shrews lack an eye field? Our inability to elicit distinct eye movements in the motor cortex of tree shrews is surprising for a number of reasons. First, tree shrews are highly visual animals (Polyak, 1957; Martin, 1990; Emmons, 2000), which is reflected in a number of visual system specializations, including a large, laminated optic tectum (Casagrande et al., 1972); a well-developed visual thalamus (Lyon et al., 2003); and multiple visual areas in cortex (Lyon et al., 1998). Given the limited ocular motility of tree shrews, however, we did not expect to identify primate-like eyefields, which are identified by the presence of smooth pursuit and saccadic eyes movements

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(Robinson and Fuchs, 1969; Bruce et al., 1985; Schlag and Schlag-Rey, 1985). Furthermore, tree shrews would not be expected to foveate (i.e., make saccades) given their less specialized retina (Samorajski et al., 1966). We did expect to identify a more simple gaze-shifting area, because similar fields have been identified in a diverse range of species, including a number of carnivores (Hardin et al., 1968; Klosovskii and Balashova, 1975; Guitton and Mandl, 1978) and rodents (Hall and Lindholm, 1974; Donoghue and Wise, 1982; Neafsey et al., 1986; Rapisarda et al., 1990). Moreover, Lende (1971) reported the presence of occasional eye movements in his surface stimulation study in tree shrews (in the rostral portion of our M2). Eye-blink movements are often elicited in cortex adjacent to eye fields (see, e.g., Stepniewska et al., 1993; Wu et al., 2000), and we did identify them in this study, but we found these responses scattered across a number of zones, including M2, 3b, and S2 (this was also reported by Lende, 1970). It may be that our stimulus parameters were insufficient to drive eye movements, given the somewhat conservative cut-off frequency selected for the present study (120 μA). Some investigators have reported that oculomotor movements require higher thresholds than skeletomotor regions (Wu et al., 2000), but most reports indicate that eye movements are elicited at relatively low thresholds, particularly in simian primates (Gould et al., 1986; Huerta et al., 1987). Another possibility is that our anesthetic agents suppressed eye movements, in that they are particularly sensitive to anesthesia level (Krieger et al., 1958), and this causes many oculomotor researchers to avoid using sedatives like xylazine. Unfortunately, the high metabolism of tree shrews requires the use of xylazine (in combination with ketamine) to maintain acceptable anesthesia levels. Even so, we have been able to elicit eye movements at low thresholds in primates anesthetized with a combination of Xyl and Ket (Stepniewska et al., 1993; Preuss et al., 1996; Fang et al., 2005), as have other researchers in carnivores (Ghosh, 1997). It is also possible that this tree shrew species (i.e., *Tupaia. belangeri*), or Scandentia in general, lacks an oculomotor field. Clearly, more evidence is needed to support any of these conclusions. The results of connectional studies currently underway may provide some clarification.

Do tree shrews lack a foot representation in motor cortex? We were less surprised by our inability to elicit hindlimb movements outside of somatosensory cortex, insofar as no corticospinal axons have been identified past the thoracic level in tree shrews (Jane et al., 1969). Moreover, similar findings have been reported in surface and intracortical stimulation studies in rodents (Hall and Lindholm, 1974; Donoghue and Wise, 1982; Neafsey et al., 1986; Rapisarda et al., 1990; Li and Waters, 1991). In rodents, this result has commonly been interpreted as representing a more primitive type of cortical organization in which the primary motor and somatosensory fields partially overlap. In rats, for example, the hindlimb motor region is said to overlap fully with the somatosensory region, because movements can be driven here at moderately low thresholds, and this region has a well-developed granular region and large layer V neurons (for review see Wise and Donoghue, 1986). An incomplete representation of the body in M1 may indeed represent a more primitive form of cortical organization, but we do not interpret these results as indicating a partial sensorimotor amalgam. It is not unusual for movements to be elicited at moderate to low thresholds within the primary somatosensory cortex of species that have a complete representation of the body in M1 (see, e.g., Fogassi et al., 1994; Preuss et al., 1997). Second, layer V pyramidal neurons are not uniform in size across either M1 or S1; their soma size scales with the distance for which they project in the spinal cord. The corticospinal neurons in the medial portions of M1 *and* S1 are generally larger than in dorsal cortex, because they project for longer distances in the spinal cord (Brodmann, 1909; von Bonin, 1949; Rivara et al., 2003). Similarly, corticospinal neurons in lateral cortex are generally smaller than those in dorsal cortex, because they do not have to project as far to influence face movement. We propose that hindlimb movements evoked in the somatosensory cortex of the present study reflect the activation of somatosensory neurons that normally modulate hindlimb responses. The activation of modulatory corticospinal neurons may also account for the forelimb movements frequently evoked from the foot region of 3b. Regardless, it appears that tree shrews lack a representation of the foot in motor cortex.

Is movement topography related to stimulus parameters? Tree shrews appear to have a very limited representation of the distal forelimb but a greatly expanded representation of the mouth compared with the results of ICMS mapping studies in other well-studied mammals (Woolsey et al., 1952; Gould et al., 1986; Neafsey et al., 1986; Li and Waters, 1991; Wu et al., 2000). We used shorter stimulus trains (commonly used in rodent studies) in a subset of cases to be certain that our parameters were appropriate for the small-brained tree shrew. In particular, we wanted to ensure that temporal summation effects were not causing large cortical networks to be recruited that would cause more complex movements to be evoked or highly represented movements to be oversampled. We found that stimulus length did not affect the maps, however; the topography and thresholds were very similar with normal and shorter stimulus trains. Moreover, the proportions of distal forelimb and tongue movements remained relatively unchanged between the different parameters. Again, it is possible that our anesthetic regimen might have influenced the representation of certain movements, but it is unclear how this would limit the expression of distal forelimb movements yet cause tongue movements to be over represented. Given that Lende (1970) reported a similar proportion of forelimb and tongue responses (using a different anesthetic regimen), we believe that our maps reflect the true representation of movement in tree shrew cortex.

Behavioral correlates of movement topography. The large representation of certain movements in the tree shrew motor cortex is likely to be reflected in their behavior. The large ratio of tongue to distal forelimb movements in these animals suggests that they depend more on fine control of the mouth than fine control of the hand. This conclusion is supported by Emmon's studies of the feeding behavior of wild tree shrews (1991, 2000). She found that, although tree shrew species differ in terms of foraging strategies, all species appear to subsist mainly on insects and small pieces of fruit. Careful observation indicated that a number of species manipulated food items primarily with their mouth, especially fruit, which they rapidly consumed by using a specialized masticatory behavior. This behavior involved chewing the fruit pulp, pressing it to the roof of the mouth, then ejecting the remaining pulp,

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which is similar to that identified in specialized fruit bats (Emmons, 2000). This masticatory pattern is complex and rapid, and is likely to require a precise control of the tongue to maximize feeding efficiency and prevent inadvertent injury. The behavioral importance of the mouth to tree shrews is also reflected in the organization of their somatosensory cortex, which has a relatively large representation of both the lips and the oral cavity but a relatively small representation of the digits (Sur et al., 1980). Relative to primates, tree shrews have limited forepaw dexterity (Martin, 1990; Sargis, 2001), and the structure of their forepaw limits their ability to grasp (Sargis, 2002). Most tree shrew species appear to use their forelimbs in tandem to hold objects (Bishop, 1962; Napier and Napier, 1967), and they appear to struggle with larger food items that require manipulation with their forepaws (Emmons, 2000). It should be noted, however, that at least one species of tree shrew (*Tupaia minor*) has been reported to have a limited grasping ability (Sargis, 2001), which is likely related to its greater level of arboreality relative to other studied species.

Motor cortex in other Euarchontoglire species

As previously noted, the primary motor cortex (M1) of tree shrews shares a number of the hallmark features of M1 identified in primates and Euarchontoglire mammals in general. Thus, we conclude that the M1 region of tree shrews is homologous to the M1 region identified widely across Euarchontoglire mammals. We propose, based on differences in architecture, corticospinal connections, and movement thresholds, that the region identified here as M2 is likely a secondary motor area. It could be argued that this region represents a rostral subdivision of M1, similar to the M1r region identified in primates (Stepniewska et al., 1993) and humans (Geyer et al., 1996). However, nonhuman primate studies report only subtle differences in thresholds and architecture between caudal M1 and M1r, yet we found large differences in thresholds between our M1 and M2 regions and clear distinctions in architecture (especially the packing density of layer V neurons). Furthermore, we identify a major difference in the density of corticospinal cells between the two motor fields in tree shrew motor cortex, but this pattern is not apparent within the M1 of any primate studied by Nudo and Masterton (1990).

The sparse pattern of corticospinal connections that we identified in M2 is more consistent with the distribution of corticospinal cells that Nudo and Masterton identified in the premotor regions of prosimian and New World primates. As with the premotor cortex of primates, tree shrew M2 requires higher thresholds to elicit movements, and has a mediolateral orientation rostral to M1. In primates, the premotor cortex has been further subdivided into at least three distinct fields (PMDc, PMDr. and PMV) based on multiple anatomical and physiological criteria (for detailed review see Geyer et al., 2000a). We found only partial support for a division of the M2 field of tree shrews into two separate fields. The dorsal portion of M2 shares many features with the PMDc region of primates, including similarities in relative location, topographic organization, architecture, and corticospinal cell distribution (Nudo and Masterton, 1990; Geyer et al., 2000). In contrast, the ventral portion of M2 shares only superficial similarities with PMV. In primates, PMV can be distinguished from PMDc by the presence of a thin but distinct granular layer, smaller pyramidal neurons, lower stimulation thresholds, representation of the face and forelimb but not the body, and separate focus of corticospinal cells that is not continuous with M1 (Nudo and Masterton, 1990; Geyer et al., 2000). Similarly to PMV, lateral M2 has lower movement thresholds and smaller pyramidal cells, but these features are generally found across M1, 3a, and 3b in tree shrews and are therefore not definitive. The more distinctive features of primate PMV are not present in lateral M2, in that forelimb movements are not systematically represented here, there is no separate focus of corticospinal cells, and there is no clear evidence of a granular layer.

Only a limited comparison of secondary motor regions can be made with other orders of mammals within the Euarchontoglire clade, because few species within this clade have been investigated in detail. A comparison of the secondary motor fields of tree shrews with those of rodents reveals few shared features. For rodents, it is generally accepted that medial agranular cortex (AGm) represents a secondary motor area (Wise and Donoghue, 1986), but this region differs from M2 in tree shrews in several respects. The AGm region in rats is located more medially and is oriented more in the rostral to caudal dimension than M2, and the topography of movements in this region is very different, being limited primarily to a central representation of the forelimb, with small medial representations for whisker and eye movements (Wang and Kurata, 1998). The distribution of corticospinal labeling is also different in rats compared with tree shrews, as tracer placement in the cervical spinal cord results in a second focus of label rostral to the location of M1 (Wise et al., 1979; Rapisarda et al., 1985; Li et al., 1990; Nudo and Masterton, 1990). Nudo and Masterton (1990) have identified this pattern of corticospinal labeling in a number of other rodent and lagomorph species, suggesting that this secondary motor field is a basic feature of Glires.

Somatosensory cortex in tree shrews

Detailed histological analysis of frontoparietal cortex, in combination with the results of corticospinal tracing, allows us to identify a number of somatosensory areas in the tree shrew parietal cortex. In addition to the previously described primary and secondary somatosensory areas, we identify some cortical fields that have not previously been described for tree shrews, including 3a, SC, and PV. In addition, we identify a number of myelin-dense ovals in lateral cortex that appear to be anatomical correlates of the hindlimb, forelimb, and mouth representations in the S2/PV region. Although a detailed analysis of these results is outside the scope of this paper, we will briefly discuss these findings in relation to previous proposals for tree shrew somatosensory cortex.

Anterior somatosensory cortex. We identify a narrow strip of cortex approximately 0.5 mm wide between M1 and primary somatosensory cortex that has both a well-developed granular layer IV and large, relatively densely packed, pyramidal neurons in layer V. This region appears to correspond to the postcentralis 2 region first identified by Zilles (1978). A similar region has been described widely for mammals, as region R in marsupials, insectivores, monotremes, and some rodents; dysgranular cortex in rats; and area 3a in primates (for detailed review see Slutsky et al., 2000; Krubitzer et al., 2004). Although

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these areas have not been systematically studied in all species, the available evidence suggests that a 3a-like area is present in all mammals and was likely inherited from a common ancestor (Krubitzer et al., 2004). We identify this region as 3a in tree shrews because of its similar shape and relative location compared with the 3a of primates. We identify, caudally to 3a, a primary somatosensory cortex in the general location reported by Sur and colleagues (1980). This region is best visualized in flattened sections processed for myelin, and it is possible to identify somatic divisions that correspond to the expected locations of the foot, forelimb, face, and mouth (Sur et al., 1980, 1981b). The primary somatosensory region of tree shrews has a somatotopic organization, relative location, and mediolateral orientation similar to those of area 3b (S1 proper) of primates and is bordered by a similar transitional region (3a). Thus, we refer to this area as *3b* to emphasize its homology with area 3b in primates (Kaas, 1983). Corticospinal neurons exhibit a patchy distribution in 3b, with the forelimb region sending sparse connections from the core but dense connections from the surrounding region. This suggests that the 3b forelimb region may be involved in modulating movements of the whole forelimb rather than individual digits.

Lateral somatosensory cortex. We identify a number of myelin-dense modules lateral to the face representation of primary somatosensory cortex that have not been previously described in tree shrews. Similar modules have been described for the lateral cortex of rodents, by using cytochrome oxidase preparations (Remple et al., 2003; Henry et al., 2006) that provide histological correlates of particular body regions in S2 and PV. Myelin-dense modules have also been identified in the lateral cortex of primates, which provide correlates for the representation of individual structures of the face and mouth in 3b (Jain et al., 2001), but comparable modules have yet to be identified in the S2 or PV region of primates. The locations of these modules in tree shrews indicate that they represent the hindlimb, forelimb, and mouth regions of S2 (Sur et al., 1981a). The results of our corticospinal tracing experiments provide partial support for this proposal, insofar as corticospinal label was identified primarily in and around the region of the forelimb module following injection into the cervical spinal cord. It should be noted that these connections appear to be more limited in tree shrews than other species (Nudo and Masterton, 1990), in that corticospinal label was not identified in the lateral cortex of all cases and was sparse when identified. Thus, this region may play less of a role in movement production in tree shrews than in other mammals. An additional division was identified lateral to S2 in coronally cut sections, which we have provisionally identified as PV based on architecture and relative position. We have mapped this region using microelectrode recordings and found a somatosensory representation of the body that forms a mirror image of S2 along distal portions of the mouth, forelimb, and hindlimb, with the head representation of PV located laterally (unpublished data). Preliminary results of cortical tracing experiments also suggest that there is an additional somatosensory representation lateral to S2 in tree shrews. Consequently, the organization of the tree shrew lateral cortex appears very similar to that of the rat (Remple et al., 2003), where the myelin-dense ovals represent the distal representations of the body for both S2 and PV, with the S2 face represented dorsally and the PV face represented ventrally. It should be noted that the orientation of PV presented here differs from that identified for primates, in that the representation of the face in tree shrew PV is not found adjacent to the face representation in 3b. It could be argued that our PV is actually a different field more similar to the ventral somatosensory (VS) field identified in a number of primates (Cusick et al., 1989; Krubitzer et al., 1995; Qi et al., 2002; Wu and Kaas, 2003) and in bats (Krubitzer and Calford, 1992). We do not favor this interpretation, because 1) there is a precedent for a similarly orientated PV region in opossums (Beck et al., 1996), rats (Remple et al., 2003), and mole rats (Henry et al., 2006) and 2) PV has been identified widely in a number of mammals and is thought to be common to mammals (Kaas, 2004), whereas VS has been identified in only one nonprimate species. Regardless, the identification of this lateral field of tree shrews as PV should be considered provisional until additional evidence is available.

Caudal somatosensory cortex. The SC region can be presumed to be somatosensory in function based on its granular architecture, relative location, and corticospinal connections. Although the precise somatosensory organization of SC is unknown, the distribution of corticospinal cells and representation of movement indicate that this region is at least roughly somatotopic, with the body represented medially and the face located more laterally. The SC field is in the same relative position as area 1 in anthropoid primates and the caudal somatosensory strip in galagos and other mammals (Slutsky et al., 2000; Kaas, 2004).

Evolutionary considerations

The results of the present study indicate that the motor and somatosensory cortices of tree shrews share more features with primates than with other Euarchontoglire animals. In particular, the orientation, relative location, architecture, and topography of M1, M2, 3a, 3b, and S2 are remarkably similar between tree shrews and primates. This finding is surprising in light of the relatively undifferentiated corticospinal system of tree shrews, which travels primarily in the dorsal funiculus and makes few if any direct connections to the motor neurons in the ventral horn (Shriver and Noback, 1967; Jane et al., 1969). A similar corticospinal organization has been identified in a diverse range of species, including rodents, lagomorphs, marsupials, and ungulates (for summary see Armand, 1982), suggesting that a similar organization might have been present in a common mammalian ancestor. Microstimulation studies in a number of these species (Donoghue and Wise, 1982; Neafsey et al., 1986; Beck et al., 1996; Frost et al., 2000) indicate that they vary greatly in terms of motor cortex complexity. Thus, the differences in motor cortex complexity identified in these species appear not to be directly related to specialization of the corticospinal system. A similar argument can be made for the evolution of the motor cortex of primates. An array of at least 10–12 motor areas has been identified in a range of well-studied primate species, including prosimians (Wu et al., 2000; Fang et al., 2005), New World monkeys (Gould et al., 1986; Preuss et al., 1996), and Old World monkeys (Geyer et al., 2000a), suggesting that a relatively complex motor cortex organization might have emerged early in primate evolution (Wu et al., 2000). The degree of specialization identified in the corticospinal system of these pri-

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mates varies greatly (for detailed review see Lemon and Griffiths, 2005), especially in terms of the number of direct (monosynaptic) connections made with motoneurons in the ventral horn. These direct connections appear to be involved in digital dexterity (Lemon and Griffiths, 2005) but are found in significant numbers only in Old World monkeys, apes, and humans, indicating that this adaptation was not likely present in early primates and might have developed relatively late in primate evolution. It may be that visuomotor integration played a more important role in the early evolution of the primate motor system (Whishaw, 2003), insofar as highly developed frontal and posterior parietal systems are found in all well-studied primates (Kaas, 2004). The organization of the motor system in tree shrews could represent a transitional form of sensorimotor organization, present in a common ancestor of tree shrews and primates. The results of connectional studies currently underway may provide additional support for this conclusion, particularly if tree shrews are found to have a well-developed frontoparietal network similar to primates. Of course, the similarities between the motor cortex of tree shrews and that of primates might reflect parallel (i.e., convergent) adaptations to a semiarboreal lifestyle. Ideally, this possibility would be tested by using a broad cladistic approach to study a range of archontan species, including additional tree shrew species, but this approach is limited by the number of species available for detailed study and by the necessary investment in resources (Kaas, 2002). Another approach for identifying possible trait convergences between primates and tree shrews is to establish whether specific traits are present in other niche-matched species, such as squirrels (Kaas, 2002). The identification of similar features in the motor cortex of squirrels might indicate that similarities between tree shrews and primates reflect convergent evolutionary processes.

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